Calibration sources for an iLQR pandemic fiscal policy model

The strongest upgrades to your current source list come from three publications: the IHME/Lancet (2022) systematic analysis of infection fatality rates across 190 countries, Liu & Rocklöv (2021) for variant-specific reproduction numbers, and the Gagnon–Kamin–Kearns (2023) GDP panel that directly challenges the Goolsbee–Syverson decomposition when the outcome is GDP rather than mobility. These sources improve OECD specificity, add wave-level variation, and sharpen the calibration of the behavioral fear channel—the three dimensions where your current citations are weakest. Below is a parameter-by-parameter assessment with full citations, recommended values, and methodological notes.

Priority 1: updated epidemiological parameters for OECD wave-by-wave calibration

Effective reproduction number ρ_θ (quarterly, 1.25–1.60)

Your current range represents effective Rt under active NPIs, not the intrinsic R0—a distinction worth making explicit. The meta-analytic consensus on intrinsic R0 for the ancestral strain converges on **2.5–3.3**, with Billah, Miah & Khan (2020, *PLOS ONE*) pooling 24 studies at R0 = **2.87** (95% CI: 2.39–3.44) and Dhungel et al. (2022, *IJERPH* 19(18): 11613)—the largest meta-analysis to date with **81 studies**—reporting R0 = **2.66** (95% CI: 2.41–2.94). REPEC +2 These supersede Liu et al. (2020), which used only 14 studies.

For variant-specific calibration, Liu & Rocklöv (2021, *Journal of Travel Medicine* 28(7): taab124) provides the cleanest decomposition: **Alpha at 1.1–1.5× ancestral** (R0 ≈ 3.1–4.2) and **Delta at mean 5.08** (range 3.2–8.0). PubMed Central Davies et al. (2021, *Science* 372: eabg3055) is the companion source for Alpha's transmissibility increase. Under OECD-level NPI regimes, these translate to effective quarterly Rt of roughly 0.8–1.6, which supports your 1.25–1.60 range as representing a moderate-to-high transmission scenario. Flaxman et al. (2020) remains valid for Rt under NPIs but should be supplemented with Brauner et al. (2021) for the 41-country cross-section. medRxiv

Methodological note to acknowledge: Flaxman et al. has been critiqued for attributing all Rt reduction to NPIs without accounting for voluntary behavioral change (Dehning et al., 2020; Sharma et al., 2021). The NPI-timing collinearity problem means Flaxman's Rt estimates conflate mandatory and voluntary distancing, which matters for your model's separation of the fear channel from φ_S.

Recommended citation strategy: Cite Dhungel et al. (2022) or Billah et al. (2020) for ancestral R0, Liu & Rocklöv (2021) for variant multipliers, and Brauner et al. (2021) for Rt under NPIs. Drop Liu et al. (2020) as superseded.

Infection fatality rate δ_θ (0.005–0.025)

The single most important upgrade available is the **IHME/COVID-19 Forecasting Team (2022), "Variation in the COVID-19 infection–fatality ratio by age, time, and geography during the pre-vaccine era," *The Lancet* 399(10334): 1469–1488.** Based on **5,131 seroprevalence surveys** across 190 countries, it is the definitive pre-vaccine IFR analysis. (PubMed Central) Key findings directly relevant to your calibration:

- Global median IFR declined from **0.466%** (April 2020) to **0.314%** (January 2021), a **33% decline** attributed to improved clinical treatment— (The Lancet)supporting time-varying δ_θ. (The Lancet)
- **High-income OECD countries had the highest IFRs** due to older populations: Portugal 2.085%, Japan 1.750%, Spain 1.710%, Greece 1.637%. (ScienceDirect) (The Lancet) The OECD-specific range is roughly **0.5%–2.0%**, mapping well to your 0.005–0.025.
- **74% of cross-country IFR variation** is explained by age structure alone. (ScienceDirect)

Levin et al. (2020, *European Journal of Epidemiology* 35(12): 1123–1138) remains essential for the age-specific IFR function (exponential relationship: 0.002% at age 10 scaling to 15% at age 85). (PubMed) Meyerowitz-Katz & Merone (2020) reported a pooled IFR of **0.68%** (95% CI: 0.53–0.82%) but with extreme heterogeneity (I^2 = 99.2%) and no age adjustment— (PubMed)cite it as an early benchmark but note the IHME/Lancet analysis supersedes it.

Contested territory: Ioannidis (2021, *European Journal of Clinical Investigation* 51(5): e13554) (PubMed Central) argues for a global IFR of ~0.15% (ResearchGate) and critiques selection criteria in Levin et al. and Meyerowitz-Katz & Merone. (Wiley Online Library) Pezzullo et al. (2023, *Environmental Research* 216: 114655) reports median IFR of **0.034% for ages 0–59**. (PubMed) These estimates are substantially lower and methodologically controversial—the Santa Clara seroprevalence study underlying several Ioannidis estimates has documented issues. (Absolutely Maybe) For an AER paper, acknowledging this debate in a footnote while anchoring calibration to the IHME/Lancet analysis is the appropriate approach.

Recommended citation strategy: Lead with IHME/COVID-19 Forecasting Team (2022, *Lancet*) as the primary source; retain Levin et al. (2020) for the age-specific functional form; note the Ioannidis critique as a lower-bound alternative.

NPI suppression effectiveness φ_S (0.4–0.7)

Brauner et al. (2021, *Science* 371: eabd9338) remains the gold standard and need not be replaced. (ResearchGate) Across 41 countries with **206 robustness checks**, they estimate that combined NPI packages reduce Rt by **60–80%**, (medRxiv) with individual interventions contributing: gathering limits ≤10 people (**42%** reduction), school/university closures (**38%**), business closures (**35%**), and stay-at-home orders (**13%**). (Science) Your φ_S = 0.4–0.7 maps directly onto these combined-NPI estimates.

Two supplementary sources strengthen the calibration. Haug et al. (2020, *Nature Human Behaviour* 4(12): 1303–1312) analyzed **6,068 NPIs across 79 territories** (Nature) and found that

combinations of less disruptive measures can match lockdown effectiveness— Semantic Scholar relevant if your model distinguishes NPI intensity levels. A 2024 rapid review in *Frontiers in Public Health* (12: 1426992) covering Italy, the US, UK, and China reported full lockdowns reducing Rt by **40–90%**, with most studies in the 70–80% range. PubMed Lison et al. (2023, *Lancet Public Health* 8(4): e311–e317) provides a methodological assessment of NPI effectiveness studies and recommends measuring effectiveness as relative changes in reproduction number—useful as a methodological justification for your modeling approach. The Lancet

The Herby–Jonung–Hanke controversy: This Johns Hopkins working paper (2022, Studies in Applied Economics No. 200) claimed lockdowns reduced COVID mortality by only **3.2%**. medRxiv It has **not been published in any peer-reviewed journal**, Science Feedback included 12 of 34 studies that were themselves working papers, used a narrow definition of "lockdown," Snopes and was criticized by Ferguson (Imperial College), multiple epidemiologists, and fact-checking organizations. Managed Healthcare Exec... For an AER submission, acknowledge its existence in a footnote while noting it measures mortality (not transmission), uses idiosyncratic inclusion criteria, and remains unpublished. Do not use it as a calibration source.

Recommended citation strategy: Retain Brauner et al. (2021, *Science*) as primary; add Haug et al. (2020, *Nature Human Behaviour*) for robustness; cite Lison et al. (2023, *Lancet Public Health*) for methodological justification; footnote the Herby et al. debate.

Priority 2: the behavioral fear channel β_d reveals a measurement-dependent puzzle

The micro-mobility consensus: fear dominates (80–90%)

Two landmark studies using within-country identification find that **voluntary behavioral responses accounted for the vast majority of the economic decline** when measured through mobility or spending. Goolsbee & Syverson (2021, *Journal of Public Economics* 193: 104311) used SafeGraph cellular data on 2.25 million US businesses with a border-discontinuity design, finding that of a **60 percentage point** decline in foot traffic, shelter-in-place orders explained only **7 pp** (~12%). SSRN The remaining **~88% was voluntary**, driven by local mortality.

Andersen, Hansen, Johannesen & Sheridan (2020, *PNAS* 117(34): 20468–20473) exploited the Denmark–Sweden natural experiment—near-identical outbreaks, starkly different policies. Aggregate spending fell **25% in Sweden** (no lockdown) versus **29% in Denmark** (lockdown), implying mandated restrictions caused only **4 additional percentage points** of decline. arXiv +2 The voluntary share: **~86%**. This Scandinavian evidence is particularly relevant to your OECD sample. ProMarket

Aum, Lee & Shin (2021, *Labour Economics* 70: 101983) provides a lower bound using Korea's localized Daegu outbreak (no lockdowns, only testing/tracing). A 1 per thousand increase in infections caused a **2–3% employment drop** in Korea, versus **5–6%** in the US/UK (with lockdowns)—implying **at least ~50%** of advanced-economy job losses were voluntary. NBER

PubMed Their companion paper in *Journal of Public Economics* (194, 2021) extends this to inequality implications. SSRN

Maloney & Taskin (2020, World Bank WP 9242) confirmed the pattern globally: in advanced economies, voluntary distancing driven by case counts explains more mobility decline than NPIs, though the relationship reverses in lower-income countries where formal enforcement matters more. SSRN IMF

The macro-GDP challenge: lockdowns dominate when the outcome is GDP

A striking discrepancy emerges when researchers use **GDP rather than mobility** as the outcome variable. König & Winkler (2021, *Intereconomics* 56(1): 32–39) ran panel regressions on **42 countries' quarterly GDP growth** with both the Oxford Stringency Index and fatality rates as regressors. Springer Springer Both were significant, but **stringency was the dominant driver** of GDP variation—the reverse of the mobility finding. PubMed Central Springer Their follow-up in *PLOS ONE* (2021) extended through Q4 2020 with similar results. PLOS

Most strikingly, **Gagnon, Kamin & Kearns (2023, *Journal of the Japanese and International Economies* 68: 101258)** analyzed **90 countries over 2020Q1–2021Q4**—the closest match to your sample's coverage and time horizon. They found COVID deaths had a **very small and sometimes statistically insignificant effect** on GDP, while lockdown stringency was the dominant driver. PubMed Central PubMed International trade emerged as a major transmission channel missed by single-country studies. PubMed Central

Chen, Igan, Pierri & Presbitero (2020, IMF WP 20/125) added institutional weight, finding **no robust evidence of additional NPI impact** beyond what mobility already captured in European and US data, while also noting that economic deterioration preceded NPI introduction and recovery began before formal reopening. IDEAS/RePEc

Reconciling the 88% vs. "small effect" puzzle

This discrepancy is not contradictory—it reflects measurement differences that matter for your calibration:

- **Mobility overstates GDP effects** because remote work maintains output while reducing foot traffic. Goolsbee and Syverson measure visits, not value-added.
- **GDP captures supply-side and trade channels** (investment, global supply chains, trade disruption) that are not driven by individual fear but correlate with NPI stringency.
- **The Oxford Stringency Index partially absorbs voluntary responses** because governments tightened measures precisely when fear was highest, creating an endogeneity that biases the voluntary coefficient downward in GDP regressions.
- **Deaths are a lagging, noisy proxy** for contemporaneous fear; excess mortality would be cleaner but is rarely used in real time.

For your iLQR model, this implies β_d should be calibrated differently depending on whether the loss function operates on mobility/consumption or GDP. If the model's activity variable is closer

to GDP, the Gagnon et al. finding suggests a **modest β_d**, with most economic damage flowing through the NPI channel. (PubMed Central) If it captures consumption or contact-intensive activity, the Goolsbee–Syverson estimate supports a **large β_d**.

The SIR-macro structural benchmark

Eichenbaum, Rebelo & Trabandt (2021, *Review of Financial Studies* 34(11): 5149–5187)—the canonical SIR-macro model— (Oxford Academic) (NBER)provides an important structural anchor. (Oxford Academic) In their model, the voluntary behavioral response alone generates a **4.7% average consumption decline** in year 1, versus only **0.7%** from a pure SIR without behavioral feedback (a **~7× multiplier**). Optimal government containment triples the recession to **22%**, (Oxford Academic) implying voluntary behavior accounts for roughly **one-fifth** of the socially optimal response. (nber) (northwestern) This structural ratio—voluntary response being substantial but well below optimal containment—offers a natural calibration target.

Recommended β_d calibration approach: Use Goolsbee & Syverson (2021) and Sheridan et al. (2020, *PNAS*) as upper-bound benchmarks (voluntary share ~85–90%), Aum et al. (2021) as the midpoint (~50%), and Gagnon et al. (2023) as the lower bound for GDP-based specifications. Report sensitivity analysis across this range. Cite the ERT (2021) structural model for the theoretical decomposition.

Priority 3: macroeconomic parameters are well-supported with minor refinements

Real interest rate r ≈ 0.001 quarterly is defensible but possibly high

Your calibration of **r = 0.001 quarterly (~0.4% annualized)** is within the plausible range but sits at the upper end of what OECD sovereign borrowing data suggest. The OECD Sovereign Borrowing Outlook 2022 (covering 2020–2021 issuance data) reports that the average nominal cost of new OECD sovereign borrowing was **1.4%** in 2021. (OECD) With OECD-average inflation of ~2–3%, the implied real rate was approximately **−0.6% to −1.6%**. US 10-year TIPS yields hovered at **−0.5% to −1.0%** throughout this period. Euro area government yields were negative in nominal terms for most maturities. (OECD)

The calibration becomes defensible when averaging across all 38 OECD members including higher-rate outliers (Turkey, Colombia, Chile), or when interpreted as a slight premium above the risk-free rate. Rachel & Summers (2019, *Brookings Papers on Economic Activity*) document that the private-sector neutral real rate for advanced economies had fallen to approximately **−2%** even pre-pandemic; (NBER) government borrowing rates were higher only because fiscal expansion partially offset this decline. (Brookings) Blanchard (2019, *American Economic Review* 109(4): 1197–1229) provides the theoretical framework for fiscal policy when r < g, directly relevant to your optimal policy analysis. (IDEAS/RePEc)

Internal consistency note: β = 0.99 implies a steady-state quarterly discount rate of ~1% (4% annualized), while r = 0.001 implies a government borrowing cost 10× lower. This wedge is empirically justified by central bank asset purchases and flight-to-safety dynamics during 2020–

2021, but should be acknowledged in the paper—the gap between the household time-preference rate and the sovereign borrowing rate is a feature, not a bug, of the pandemic fiscal environment.

Recommended sources: OECD (2022), *Sovereign Borrowing Outlook 2022* for the direct data; Rachel & Summers (2019, BPEA) for secular context; Blanchard (2019, AER) for the r < g framework. An alternative data source is the World Bank International Financial Statistics real interest rate series, which covers all OECD members. worldbank

Discount factor β = 0.99 needs no updating

This is the canonical quarterly calibration used across virtually all DSGE and pandemic-macro models. Eichenbaum, Rebelo & Trabandt (2021, *RFS*) use an equivalent value; Oxford Academic +3 Christiano, Eichenbaum & Evans (2005, *JPE* 113(1): 1–45) IDEAS/RePEc and Smets & Wouters (2007, *AER* 97(3): 586–606) establish the standard. No pandemic-macro paper has argued for deviating from β = 0.99. A single sentence citing ERT (2021) and one standard DSGE reference suffices.

Health expenditure c_H = 0.02 finds support in OECD data

The OECD *Health at a Glance 2023* reports that health spending as a share of GDP jumped by approximately **1 percentage point** across the OECD average in 2020 (from ~8.8% to ~9.8–10%), OECD with the OECD (2024) *Fiscal Sustainability of Health Systems* report specifying that **COVID-specific spending peaked at 9% of total health spending** in 2021— OECD implying COVID-specific fiscal health costs of roughly **0.9% of GDP**. The IMF Fiscal Monitor (April 2022) documents that advanced economies deployed **8–10% of GDP** in total above-the-line fiscal measures, though only a fraction was health-specific. World Bank +2

Your c_H = 0.02 (2% of GDP per unit infection) is plausible under the interpretation that each percentage point of the population infected generates ~0.02 percentage points of GDP in additional health spending. With cumulative infection rates reaching **30–50%** of OECD populations by early 2022 (accounting for unreported cases), the implied total health cost would be **0.6–1.0% of GDP**, broadly consistent with observed data. Cutler & Summers (2020, *JAMA* 324(15): 1495–1496) provide the broader cost framework—their **$16 trillion** US pandemic cost estimate (90% of annual GDP) includes lost output, premature death, long COVID, and mental health, far exceeding direct fiscal health costs. Harvard Kennedy School PubMed Central

Recommended sources: OECD (2023), *Health at a Glance 2023* for direct cross-country health spending data; IMF (2022), *Fiscal Monitor: Fiscal Policy from Pandemic to War* for fiscal response decomposition; Cutler & Summers (2020, *JAMA*) for the comprehensive cost framework.

Conclusion: three actionable upgrades and one unresolved tension

The most consequential upgrade is replacing Meyerowitz-Katz & Merone (2020) and supplementing Levin et al. (2020) with the **IHME/Lancet (2022) IFR analysis**, which provides OECD-specific, time-varying estimates from 5,131 seroprevalence surveys—exactly the kind of

source an AER referee would expect. The second upgrade is adding **Liu & Rocklöv (2021)** and **Davies et al. (2021, *Science*)** for variant-specific reproduction numbers to support wave-by-wave calibration. The third is citing **Haug et al. (2020, *Nature Human Behaviour*)** alongside Brauner et al. to strengthen the NPI effectiveness range.

The deepest unresolved tension in your calibration concerns β_d. Micro-level studies (Goolsbee–Syverson, Sheridan et al.) attribute **85–90%** of the economic decline to voluntary fear, PubMed +2 while macro-GDP panels (Gagnon et al., König–Winkler) find lockdown stringency dominates. Springer PubMed Central This is not a data quality issue but a genuine measurement discrepancy: mobility ≠ GDP, and the Oxford Stringency Index absorbs endogenous policy responses to fear. For an AER paper, the strongest approach is to **present β_d calibrations spanning both ends of this range** and show that your optimal policy conclusions are robust. The ERT (2021) structural model—where voluntary behavior generates one-fifth of the optimal containment effect—offers a natural midpoint anchor. nber +2

Parameter	Value	Primary Source (New/Updated)	Secondary Source	Key Upgrade
ρ_θ (Rt)	1.25–1.60	Brauner et al. (2021, *Science*)	Liu & Rocklöv (2021, *J Travel Med*); Dhungel et al. (2022, *IJERPH*)	Variant-specific R0 multipliers
δ_θ (IFR)	0.005–0.025	**IHME/Lancet (2022)**	Levin et al. (2020, *Eur J Epidemiol*)	190-country seroprevalence; time-varying IFR
φ_S (NPI)	0.4–0.7	Brauner et al. (2021, *Science*)	Haug et al. (2020, *Nat Hum Behav*); Lison et al. (2023, *Lancet PH*)	Methodological robustness
β_d (fear)	Range	Goolsbee & Syverson (2021, *JPubE*)	Sheridan et al. (2020, *PNAS*); Gagnon et al. (2023, *JJIE*); ERT (2021, *RFS*)	GDP-vs-mobility reconciliation
r (rate)	0.001/qtr	OECD Sovereign Borrowing Outlook 2022	Rachel & Summers (2019, *BPEA*); Blanchard (2019, *AER*)	r < g framework
β (discount)	0.99	ERT (2021, *RFS*); CEE (2005, *JPE*)	—	No change needed
c_H (health)	0.02	OECD Health at a Glance 2023	IMF Fiscal Monitor (Apr 2022); Cutler & Summers (2020, *JAMA*)	OECD-specific health spending data